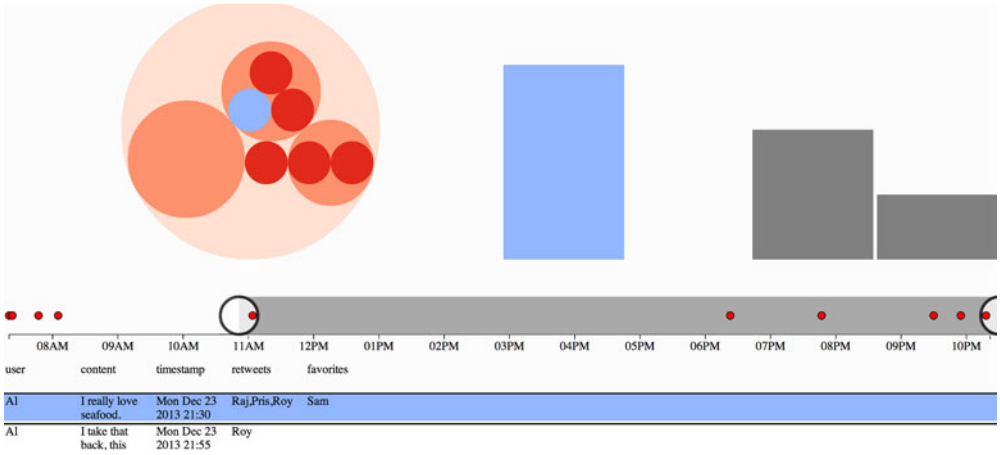


Figure 9.1 Throughout this chapter, we'll build toward this fully operational data dashboard, first creating the individual chart elements (section 9.1), then adding interactivity (section 9.2), and finally adding a brush to filter the data by time (section 9.3).

Infoviz term: dashboard

Multiple charts combined into a single application have been around since the 1970s and were traditionally associated with decision support systems. Dashboards provide the kind of multiple views into a dataset that you'll see in this chapter, and are often the selling point of charting libraries like NVD3.

Although they're typically presented as several charts sharing screen space, the principles of data dashboards can also be applied to web mapping and text-based applications through modal pop-ups or any website that provides several different charts simultaneously. In those cases, the act of highlighting datapoints may be a response to the scrolling of text or zooming in on a map, rather than mousing over a data visualization element.

To get started, we need to design the interface that we want. Designs can be rough sketches or detailed sets of user requirements. Figure 9.2 shows a simple sketch that combines several of the charts we used in previous chapters. Each of these could represent the same dataset from tweets.json, but in a different manner. With a data dashboard, we want to provide a user with multiple perspectives of the data as well as the ability to drill down into the data and see individual datapoints. We'll use a circle pack like we used in chapter 5 so that users can differentiate tweeters and their tweets, a bar chart for raw number of tweets, and a spreadsheet like the one in chapter 8 so that users can view the content of individual tweets. We also want to let users slice and dice their data, so later we'll add that functionality with a brush. Both the bar chart and circle packing require that we use `d3.nest()`, whereas the spreadsheet takes the unchanged object array.